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Skin-Tone Steering at Matched Change: A Preregistered Evaluation of Denoising-Time Latent Control in SDXL

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Abstract

We study whether a linear direction estimated from paired portrait latents can control visible skin tone in Stable Diffusion XL while preserving other image properties. The proposed method computes a paired mean difference in the VAE latent space, applies a spatial Gaussian mask, and distributes the resulting perturbation across denoising steps. We preregister a matched-change evaluation after calibration, estimate the direction from 64 of 96 noise-coupled prompt pairs, validate an image-colour instrument on the remaining 32, and evaluate 570 conditions over 30 held-out seeds. The parent study found no detectable face-embedding advantage, but masking reduced LPIPS relative to unmasked steering by 0.023 lighter and 0.082 darker (Holm-adjusted $p = .012, .002$). We then prospectively fixed an independent-seed replication with new direction data and 30 new seeds. Its LPIPS advantages were again positive: 0.024 ($[0.005, 0.041]$) lighter and 0.041 ($[0.018, 0.065]$) darker. The locked decision was nevertheless inconclusive: only 8 and 10 shared matched seeds were available, below the minimum of 12, and the lighter two-test Holm value was .056 (darker .019). The cross-campaign masked-direction cosine was 0.919. The evidence supports a localized perceptual-preservation signal under this SDXL protocol, not strict replication, identity preservation, disentanglement, or bias mitigation. Skin tone is treated only as an apparent visual attribute, never as a proxy for race or ethnicity.

1. Introduction

Latent diffusion models synthesize high-resolution images by denoising in a compressed representation rather than directly in pixel space [7]. SDXL extends this family with a larger U-Net and richer conditioning [6]. These models offer expressive control but also make it difficult to isolate one visual attribute without changing identity, pose, or composition.

Linear semantic directions have been effective in GAN latent spaces [9]. We ask whether an analogous direction, estimated from paired SDXL VAE encodings, can be used during diffusion sampling. Our technical hypothesis is that a denoiser can absorb small perturbations injected throughout sampling more coherently than a single post-hoc edit. Our scientific hypothesis is narrower: at a matched change in measured skin tone, stepwise masked injection will preserve non-target properties better than relevant baselines.

We make three contributions. First, we specify the paired estimator and denoising-time intervention. Second, we provide a preregistered matched-change comparison and a prospectively specified independent-seed replication that separate target response from preservation. Third, we release an auditable research artifact with immutable model and data hashes, measurement validation, seed-level uncertainty, missingness, and failures.

2. Related work

This study does not claim that semantic directions in diffusion models are new. Asyrp identifies a semantic representation space inside frozen diffusion U-Nets and manipulates directions across the reverse process [3]; later work discovers concept-specific internal directions for text-to-image generation [4]. Concept Sliders instead learn low-rank parameter directions for continuous, composable control [1]. Our narrower object of study is a direction formed directly from paired SDXL VAE encodings, added to scheduler latents throughout sampling without optimization. The empirical question is whether that simple intervention offers a preservation advantage at matched target response, not whether diffusion representations admit semantic control in general.

3. Construct and scope

Race and ethnicity are social identities, not one-dimensional visual attributes. The intervention in this study is therefore called a skin-tone direction. It must not be interpreted as changing, estimating, or classifying race. Even datasets designed to balance face categories expose the difficulty and consequences of reducing social categories to visual labels [2]. The present method is intended for controlled generative-model research and auditing, not profiling or decisions about people.

4. Method

Let $E(x)$ be the deterministic posterior-mode encoding of an image under the SDXL VAE. Given n noise-coupled prompt pairs (x_i^L, x_i^D) generated with the same seed and prompt template but different skin-tone descriptors, the raw direction is

$$v = \frac{1}{n} \sum_{i=1}^{n} \left[E(x_i^D) - E(x_i^L) \right]. \tag{1}$$

Seed reuse controls the initial diffusion noise but does not guarantee the same apparent identity after conditioning changes. Identity, composition, prompt wording, and illumination therefore remain potential training-direction confounders.

We multiply v by a spatial mask $M \in [0,1]^{H \times W}$ centered on the face:

$$M(p) = w_e + (w_c - w_e) \exp\left[-3 \left(\frac{\|p\|_2}{r} \right)^2 \right], \quad \tilde{v} = M \odot v, \tag{2}$$

where the study uses center weight $w_c = 1$, edge weight $w_e = 0.3$, and radius $r = 1$. For T reverse-diffusion steps and steering magnitude α, the pipeline updates the scheduler latent after each step by

$$z_t \leftarrow z_t + \frac{\alpha}{T} \tilde{v}. \tag{3}$$

The implementation uses 25 DPM-Solver++ multistep updates, motivated by fast diffusion ODE solvers [5], with classifier-free guidance 7.5.

5. Evaluation design

The target response and preservation outcomes must be measured separately. The continuous target proxy uses cheek-pixel CIE Lab individual typology angle (ITA). Instrumental ITA decreases as measured pigmentation increases [12], but apparent skin color in images is affected by hue, illumination, and acquisition conditions and is not fully represented by a single light–dark axis [10]. We therefore validate ordering and prespecified illumination perturbations on a held-out split and describe the result only as apparent image color.

The primary preservation outcome is face-embedding similarity, following the general embedding approach of FaceNet [8], at a matched continuous skin-tone change. Secondary outcomes are LPIPS perceptual distance [13], background-only SSIM [11], head-pose drift, face-detection failure, and monotonicity.

The confirmatory design used held-out seeds and four methods: prompt-only generation, post-hoc latent addition, unmasked stepwise injection, and masked stepwise injection. The preregistered configuration contains 30 held-out generation seeds and method-specific grids frozen after calibration. Comparisons are paired within seed. We report means, standard deviations, medians, detector failures, and 95% seed-level bootstrap confidence intervals; secondary confirmatory tests use Holm correction. A run cannot pass when a required metric is missing.

After fixing the parent result, we prospectively specified an independent-seed replication with a newly generated 96-pair campaign, 64 pairs for a new direction, 32 for the unchanged validation gates, and evaluation seeds 600000–600029. The model revision, method grids, target, tolerance, outcomes, and bootstrap procedure remained fixed. The locked replication family contains only the lighter- and darker-direction masked-versus-unmasked LPIPS contrasts. Strict replication requires both positive estimates, both ordinary 95% bootstrap intervals above zero, and both two-test Holm-adjusted $p < .05$; either nonpositive estimate is failure, while fewer than 12 shared matched seeds in either direction is inconclusive. Parent and replication samples are reported separately rather than pooled as a 60-seed confirmatory study.

6. Pre-confirmatory validation and calibration

The direction-data campaign generated 96 noise-coupled light/dark prompt pairs. We reserve 64 pairs for direction estimation and 32 pairs for measurement validation. The ITA instrument detected the face and cheek regions in all 32 held-out pairs, ordered the light-descriptor image above the dark-descriptor image in 31 of 32 pairs (96.9%), and produced a median within-pair separation of 16.67 ITA degrees. Across prespecified brightness and white-balance perturbations, the median absolute ITA shift was 0.40 degrees and the 95th percentile was 3.06 degrees. Table 1 shows that every frozen gate passed before confirmatory generation.

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Table 1. Held-out validation of the continuous image-colour instrument.

Criterion	Required	Observed
Detection rate	≥ 0.95	1.000
Pair-order accuracy	≥ 0.90	0.969
Median pair gap (ITA)	≥ 8.0	16.67
Median perturbation shift	≤ 2.0	0.40
95th-percentile shift	≤ 5.0	3.06

Calibration used seeds disjoint from both the direction-data and confirmatory sets. The frozen target was an absolute five-degree ITA change with a three-degree matching tolerance. Prompt-only, unmasked stepwise, and masked stepwise steering achieved both directions for all three calibration seeds. Post-hoc addition achieved the darker target for all three seeds but the lighter target for only one; wider endpoint probes remained non-monotonic and identity-dependent. We therefore froze post-hoc addition as a descriptive feasibility and dose-response baseline, excluding it from the primary matched contrast before confirmatory generation.

7. Confirmatory results

7.1. Completion and target response

The runner produced all 570 prespecified condition rows with 570 unique keys and the frozen configuration fingerprint. All images generated. One post-hoc condition (seed 500019, $\alpha = -1.5$) lacked a detectable edited face/skin region, leaving 569 metric-complete rows (99.8%). The failure remains in the missingness denominator. All 330 conditions in the primary matched-change methods were complete.

Figure 1 shows the mean target response. Mean seed-level Spearman correlations between α and skin-tone change were 0.95 for prompt-only, 0.93 for unmasked stepwise, 0.92 for masked stepwise, and 0.67 for post-hoc addition. The post-hoc median was 0.99 but its mean 95% interval was $[0.43, 0.88]$, confirming substantial seed dependence.

The target was reached within the three-degree tolerance unevenly (Table 2). Masked steering had the highest coverage in both directions. Among successful matches, its mean achieved changes were 5.06 ITA lighter and 5.23 ITA darker. Low prompt-only lighter coverage constrains that stratum to six seeds shared with the masked reference.

Table 2. Prespecified matched-target coverage over 30 seeds.

Method	Lighter	Darker
Prompt-only	7 (23%)	18 (60%)
Stepwise, unmasked	13 (43%)	15 (50%)
Stepwise, masked	19 (63%)	26 (87%)

7.2. Preservation at matched change

The primary face-similarity hypothesis was not supported. Masked-stepwise advantages relative to prompt-only were -0.0008 (95% CI $[-0.0035, 0.0016]$, $n = 6$) for lighter edits and 0.0032 ($[-0.0056, 0.0149]$, $n = 17$) for darker edits. Relative to unmasked stepwise steering, the advantages were 0.0020 ($[-0.0014, 0.0056]$, $n = 13$) and 0.0079 ($[-0.0027, 0.0184]$, $n = 14$). All Holm-adjusted p values were 1.0.

Masking nevertheless improved perceptual preservation relative to unmasked steering: the paired LPIPS advantage was 0.0235 ($[0.0128, 0.0360]$, $p_{\text{Holm}} = .012$) for lighter edits and 0.0825 ($[0.0476, 0.1302]$, $p_{\text{Holm}} = .002$) for darker edits. Against prompt-only generation, the darker-direction LPIPS advantage was 0.0812 ($[0.0379, 0.1218]$, $p_{\text{Holm}} = .028$). The corresponding darker-direction improvements were 0.0207 in background SSIM ($[0.0105, 0.0319]$, $p_{\text{Holm}} < .001$) and 1.61 in the pose-difference proxy ($[0.88, 2.36]$, $p_{\text{Holm}} = .007$). Masking also improved background SSIM by 0.0129 versus unmasked darker edits ($[0.0080, 0.0176]$, $p_{\text{Holm}} = .002$). No other secondary contrast survived Holm correction. Figure 2 shows every prespecified matched preservation contrast, including null and adverse estimates.

Figure 3 is illustrative rather than inferential. Its seed was selected by a fixed rule: maximize the number of complete matched rows, then choose the seed closest to the median base ITA, breaking ties by lower seed. The prompt-only example visibly demonstrates why matched image-colour change does not imply matched identity or composition.

8. Post-confirmatory robustness

We performed two explicitly exploratory checks after completing the frozen analysis. First, we repeated matching with one- and two-degree ITA tolerances. The LPIPS advantage retained the same positive direction in every estimable stratum, and face-similarity estimates remained compatible with no advantage. However, shared matched samples collapsed to $n = 1$–5 at one degree and $n = 3$–12 at two degrees; no secondary

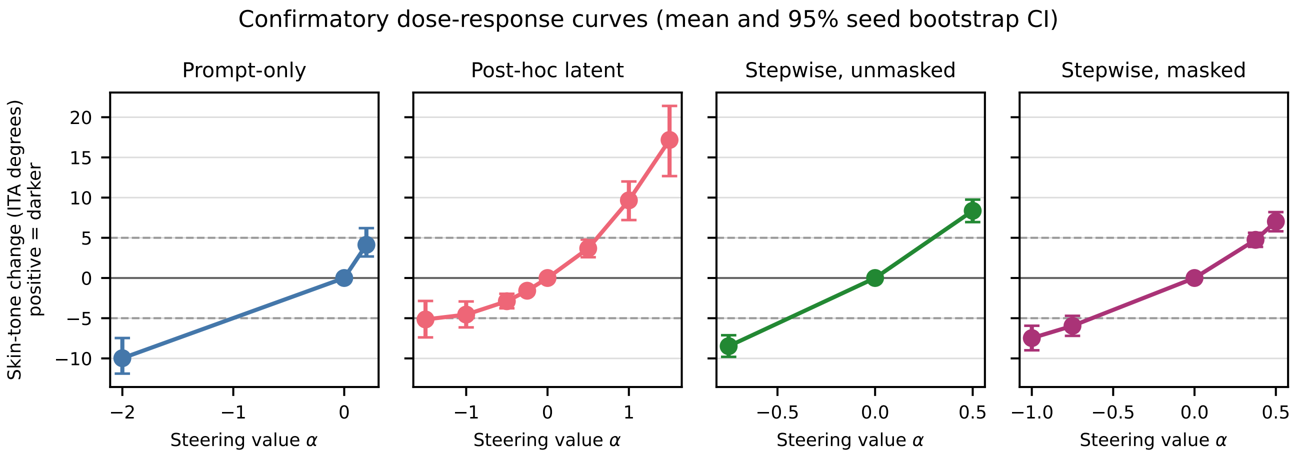

Figure 1. Confirmatory dose-response curves. Points are seed means and bars are 95% seed-level bootstrap intervals. Dashed lines mark the prespecified ±5 ITA targets; positive change denotes a darker measured appearance.

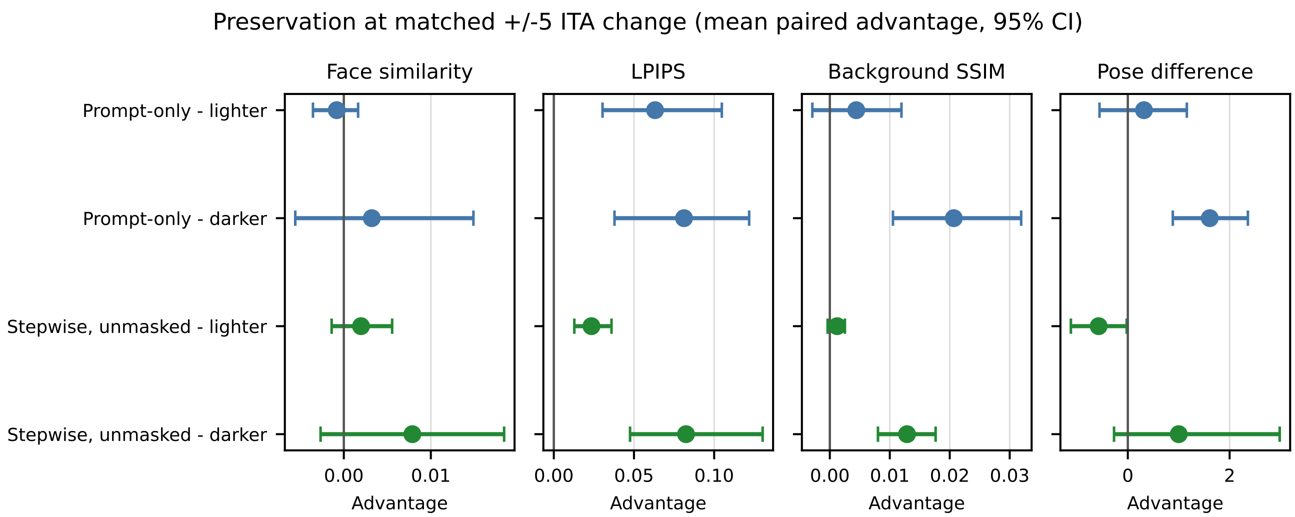

Figure 2. Masked-stepwise preservation advantage at matched ±5 ITA change. Bars are 95% seed-level bootstrap intervals. Positive values favor masked steering after orienting every outcome so that larger is better.

contrast survived Holm correction. Thus, the preregistered three-degree results should not be interpreted as evidence of precisely matched effects under stricter tolerances.

Second, we re-encoded all 64 training pairs and confirmed that the resulting direction was exactly equal to the recorded confirmatory tensor (maximum absolute difference zero). We then drew 200 random disjoint split-halves at each pair count. Table 3 reports cosine agreement between independently estimated directions. Stability increased with sample size, and masking improved agreement, but even two disjoint 32-pair estimates had median cosine 0.75 before masking and 0.82 after masking. The direction is therefore reproducible under the fixed 64-pair campaign but not sample-invariant.

Table 3. Exploratory split-half direction stability. Entries are median cosine similarity with 2.5th–97.5th percentiles over 200 disjoint resamples.

Pairs/half	Raw direction	Masked direction
8	0.42 [0.28, 0.53]	0.53 [0.38, 0.63]
16	0.59 [0.50, 0.66]	0.69 [0.60, 0.75]
32	0.75 [0.69, 0.78]	0.82 [0.77, 0.84]

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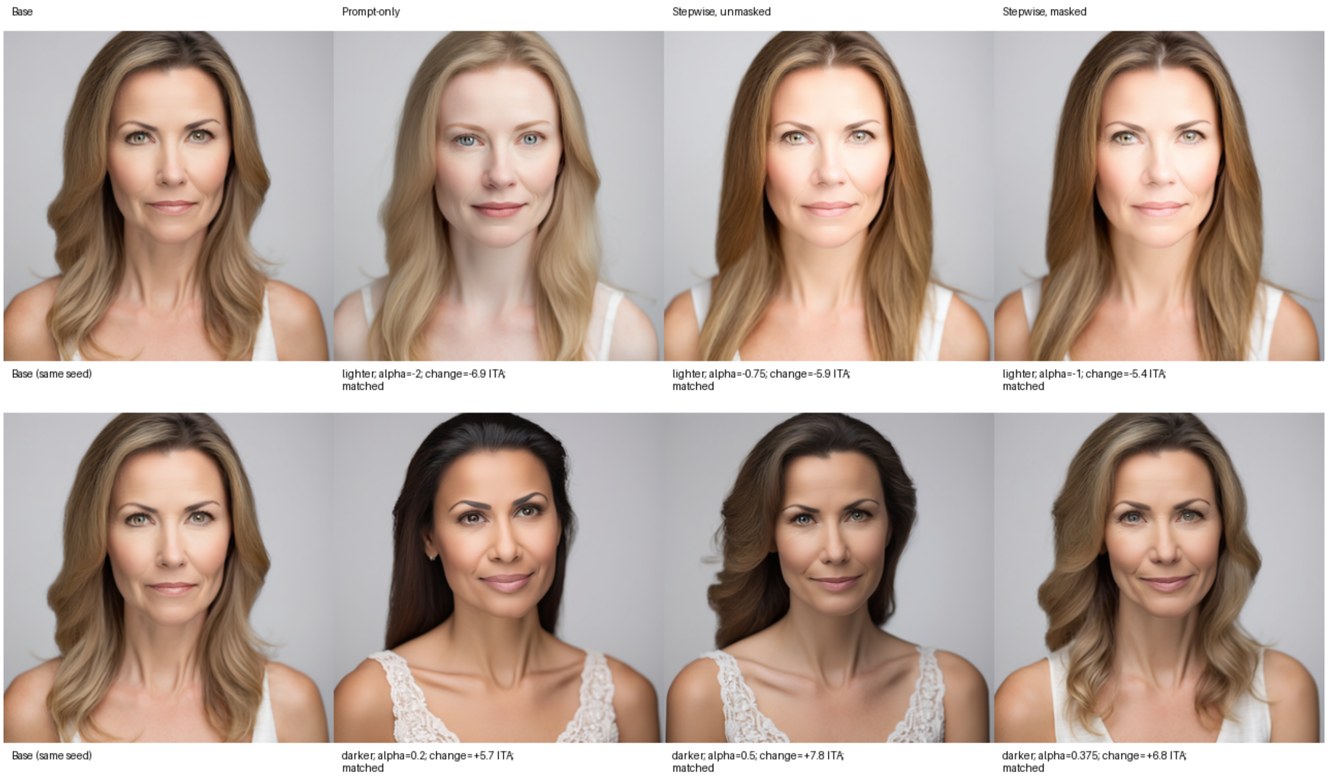

Figure 3. Deterministically selected confirmatory example (seed 500011). Each row reuses the same base image; labels report the selected alpha and achieved ITA change. Prompt-only conditioning can alter apparent identity despite reusing the diffusion seed.

9. Prospective independent-seed replication

A fresh 96-pair campaign passed the unchanged held-out measurement gates before outcome generation: detection was 1.000, pair ordering 0.969, the median pair gap 16.08 ITA degrees, and median/95th-percentile perturbation shifts 0.35/3.83 degrees. The new outcome run then completed all 570 frozen conditions over 30 new seeds. All images generated; one post-hoc condition (seed 600020, $\alpha = -1.5$) retained a face/skin-region measurement failure, leaving 569 metric-complete rows and every primary matched method complete.

Masked target coverage was 18/30 lighter and 27/30 darker, versus 10/30 in each direction for unmasked steering and 7/30 lighter and 22/30 darker for prompt-only. The locked masked-versus-unmasked LPIPS effects retained the parent signs and had ordinary bootstrap intervals above zero (Table 4). The formal decision was nevertheless inconclusive coverage: only 8 lighter and 10 darker seeds were shared between matched methods, below the prespecified minimum of 12. Even without that override, the lighter two-test Holm value of .056 missed the strict rule; the darker value was .019. We therefore report a sign-consistent, interval-positive pattern, not strict or directional replication.

Table 4. Locked independent-seed replication family. Advantage is unmasked LPIPS minus masked LPIPS; positive values favor masking. $p_{H(2)}$ is Holm adjusted across these two tests.

Direction	n	Advantage	95% CI	$p_{H(2)}$
Lighter	8	0.0238	[0.0049, 0.0415]	.0555
Darker	10	0.0409	[0.0176, 0.0655]	.0192

The independently estimated raw directions had cosine 0.884; applying the frozen spatial mask raised cross-campaign cosine to 0.919. Their norm ratio was 1.04. Figure 4 compares effects and target coverage without pooling the two campaigns.

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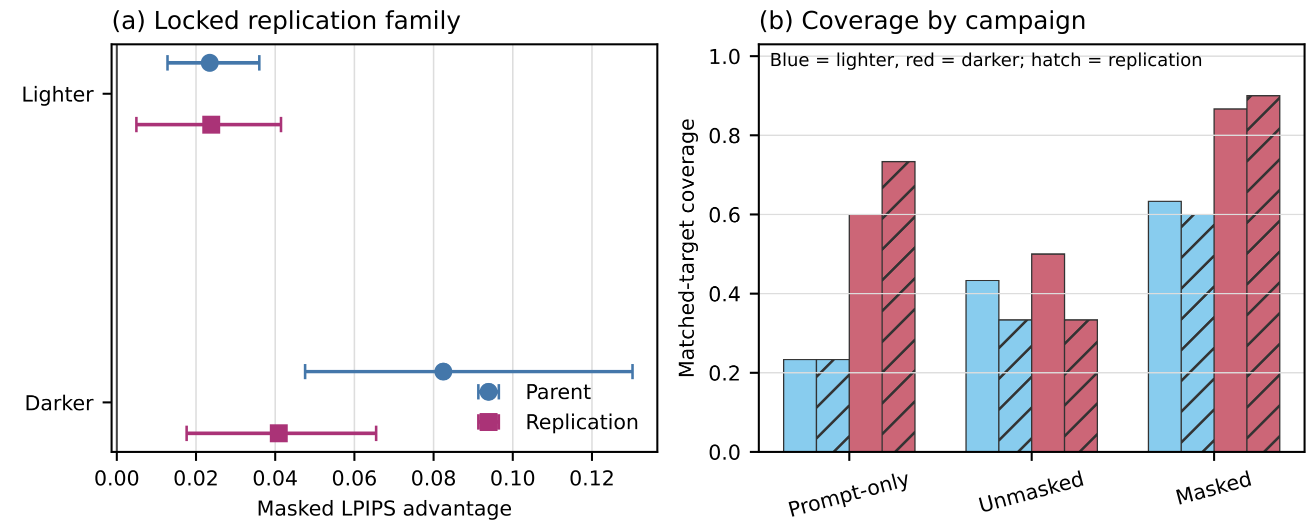

Figure 4. Parent and prospective replication reported separately. (a) Locked masked-versus-unmasked LPIPS advantages with ordinary 95% bootstrap intervals. (b) Matched-target coverage; hatching marks replication bars. The formal replication decision is inconclusive because shared matched coverage fell below 12 in both directions.

10. Limitations

The study evaluates one generator, scheduler, resolution, and model revision; cross-model generalization is unknown. Synthetic direction pairs can encode prompt wording, brightness, age, gender presentation, hair, or facial features alongside skin tone, and seed reuse does not preserve identity. A center Gaussian is not a semantic skin mask. The same generator creates direction and evaluation images, so model-specific artifacts may dominate. ITA is an image-colour proxy, not a complete representation of skin appearance, and its validation uses synthetic prompt pairs rather than calibrated physical colour measurements. Face embeddings and face/skin detectors can exhibit demographic error. Matched-change coverage is incomplete and particularly low for prompt-only lighter edits, reducing paired sample sizes and precision. The post-confirmatory tolerance analysis confirms that stricter matching is too sparse for stable inference, while the split-half analysis shows that the learned direction remains sensitive to which synthetic pairs are sampled. The prospective replication independently reproduced the effect signs and direction geometry but remained formally inconclusive because shared matched coverage was only 8–10 seeds. Both campaigns use the same generator, prompt template, descriptors, and calibration grids, so this is seed-level rather than model- or construct-level independence. The closest-grid matching rule also selects on a noisy measured response; denser grids or prespecified dose–response interpolation would reduce that source of selection and residual mismatch. The pose measure is a five-landmark proxy rather than a full 3D estimator. No human study assesses perceived identity, realism, or social harms. Finally, the additive update is scheduler-dependent and has not been validated across model or library revisions.

11. Conclusion

In the parent campaign, masked denoising-time injection offered higher target coverage than the tested unmasked grid and improved LPIPS preservation in both directions. A prospectively specified new-seed campaign produced smaller but same-sign LPIPS advantages with intervals above zero and high cross-campaign direction cosine. Its locked decision was inconclusive because only 8–10 shared matched seeds were available, and the lighter multiplicity-adjusted test narrowly missed .05. The preregistered face-similarity hypothesis was not supported in either campaign's broader corrected family. The result must not be described as strict replication, identity preservation, disentanglement, or bias mitigation. The principal contribution is a bounded empirical signal and an auditable protocol for separating target response, preservation, missingness, and unsupported social claims.

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